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Adapting Image Classification Adversarial Detection Methods for Traffic Sign Classification in Autonomous Vehicles: A Comparative Study

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***Abstract*— Autonomous driving systems critically depend on accurately classifying traffic signs, a task that is jeopardized by adversarial attacks. This paper focuses on the relatively unexplored domain of Traffic Sign Classification (TSC) in the context of detecting adversarial attacks. We conduct a rigorous evaluation of six prominent adversarial example detection methods, each representing a distinct detection category and well-regarded within the research community. Our methodology involves adapting these techniques, originally developed for general image classification (IC), to the unique challenges posed by traffic sign images, characterized by their complexity due to factors like varying environmental conditions and a large number of classes. Our study reveals insights into their effectiveness, with the Natural Scene Statistics (NSS) method outperforming others with 83.42%, 86.42%, and 99.96% detection rates; 0.08%, 0.05%, and 0.04%, false positive rates; and 0.02sec, 0.01sec, and 0.02sec detection times for Chinese, Belgium and German traffic sign datasets, respectively. NSS's superiority is crucial for autonomous vehicles. Our study also sheds light on the often-neglected aspect of detection time in IC, which plays a vital role in ensuring operational efficiency and safety for autonomous vehicles. Our research highlights the need for customized defense strategies tailored to the TSC domain, considering our evaluation's findings. By identifying promising techniques for detecting adversarial attacks in TSC, we contribute to enhancing the safety and robustness of autonomous driving systems. This study fills a critical knowledge gap, providing valuable insights into understanding and defending against adversarial attacks, specifically in the TSC context.**

***Index Terms*— Adversarial attacks, autonomous vehicles, detection methods, traffic sign classification.**

I. Introduction

AUTONOMOUS VEHICLES (AVs) are reshaping the transportation landscape, offering the promise of safer and more efficient journeys [1]. At the heart of AV development lies Traffic Sign Recognition (TSR), a vital component that ensures the safe and effective operation of these vehicles [2]. TSR encompasses several stages, starting with the acquisition of traffic sign images, followed by preprocessing, traffic sign detection (TSD), and ultimately, traffic sign classification (TSC) [3].

The significance of traffic signs cannot be overstated, as they play a fundamental role in guiding AVs through intricate road networks with both safety and efficiency [4]. TSD is responsible for identifying the presence and precise locations of traffic signs within the acquired images. By accurately pinpointing these visual cues, TSD enables AVs to recognize and interpret traffic signs effectively [5]. Once the traffic signs are detected, TSC takes over, categorizing them into specific classes such as school zone, railway crossing, speed limit to 80km/hr, to mention a few. The accuracy of this classification directly impacts the decision-making and response mechanisms of AVs, ensuring they interpret and respond appropriately to the conveyed information [5].

Unlike some perception tasks that can rely on a fusion of sensor data such as obstacle detection or lane tracking and detection, traffic sign recognition predominantly hinges on data from camera sensors [5]. The camera lens becomes the AV's eye, capturing images of traffic signs and relaying them to deep neural networks (DNNs) for interpretation and decision-making [6]. These DNNs, especially Convolutional Neural Networks (CNNs), have demonstrated exceptional prowess in recognizing traffic signs under normal conditions, attaining levels of accuracy that surpass human performance [7], [8], [9], [10], [11], [12], [13].

However, the integration of DNNs into TSC has also introduced a potential vulnerability: adversarial attacks (AAs). These attacks involve manipulating traffic sign images to create "adversarial examples" (AEs), these are modified traffic signs that deceive DNNs. This could lead to misinterpretation of traffic signs, potentially resulting in hazardous decisions for AVs. This critical situation of vulnerability, first highlighted by Szegedy et al. [14], underscores the urgent need for robust and accurate TSC systems to ensure the safety and reliability of AV navigation. Moreover, given TSC's reliance on visual data

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and the absence of supplementary sensor inputs for validation, detecting AEs is paramount.

Recent advancements in sensing technologies, such as the vision-based soft tactile sensing system presented in ViART [15], underscore the potential for enhancing detection methods in AVs. ViART's robust environmental sensing and real-time processing capabilities emphasize the importance of integrating advanced sensing modalities to improve the detection and mitigation of adversarial examples.

Similarly, real-time intelligent detection systems like AutoDetect [16] demonstrate the potential for enhancing detection methods in TSC. AutoDetect's real-time processing and intelligent sensor failure detection capabilities highlight the need for incorporating advanced detection modalities to improve the identification and mitigation of adversarial threats.

The complexity of AAs in TSC is evident in the diverse array of sophisticated methods employed to undermine classification models in AV systems. For instance, Sitawarin et al. [17] utilized the Carlini & Wagner (CW) attack method and Expectation over Transformation (EoT) to subtly alter traffic signs, effectively misleading TSC models. Ye et al. [18] developed an adversarial patch technique that deceives DNNs into misclassifying traffic signs, highlighting a significant security risk in AV technology. Moreover, Sitawarin et al. [19] introduced the DARTS strategy, which manipulates adversarial traffic signs to appear differently based on the viewing angle, challenging even the most robust systems. Additionally, Liu et al. [20] employed generative adversarial networks (GANs) with attention mechanisms to craft deceptive stickers for traffic signs, strategically placed to deceive TSC models. These varied and evolving adversarial attack generation methods in TSC highlight the need for advanced detection-based defense strategies in autonomous driving technology.

While adversarial attack detection in general image classification is well-researched [21], [22], [23], the challenges specific to TSC, remain under-explored. Therefore, this study presents a comprehensive comparative evaluation of general image classification (IC) detection techniques adapted specifically for TSC against white-box adversarial attacks. Our methodology involves evaluating six distinct adversarial detection methods: Natural Scene Statistics (NSS) [24], Local Intrinsic Dimensionality (LID) [25], Selective and Feature-based Adversarial Detection (SFAD) [26], Deep Neural Rejection (DNR) [27], Feature Squeezing (FS) [28], and Adaptive Noise Reduction (ANR) [29]. Each of these methods offers a unique perspective in identifying adversarial perturbations.

Additionally, the analysis was conducted on diverse traffic sign image datasets: the Chinese Traffic Sign Recognition Database (TSRD) [30], Belgium Traffic Sign Classification (BTSC) [31], and the German Traffic Sign Recognition Benchmark (GTSRB) [32]). These datasets not only present unique challenges due to real-world perturbations and the safety implications of misclassifications but are also compounded by a myriad of factors. These include varying environmental conditions such as illumination changes and weather effects, physical obstructions like occlusions, the impact of aging on sign visibility, human-induced alterations, and the complexities arising from strong perspectives and diverse aspect ratios of traffic signs. Each of these elements adds layers of complexity to the task of accurately classifying traffic signs, underscoring the need for robust and adaptable adversarial attack detection methods in TSC.

A. Contributions

The contributions of our work are summarized below;

1) To the best of the authors' knowledge, this is the first study to introduce the adaptation of other domains' established detection methods against white-box attacks specifically to the detection of adversarial examples for traffic sign images, representing a significant advancement in the application of proven IC techniques to the domain of TSC.
2) A thorough evaluation of these adapted methods is conducted on diverse TSC datasets, enriching the understanding of TSC complexities and assessing the robustness of systems against AEs. This comprehensive analysis establishes a new benchmark for future research in the field of TSC.
3) Acknowledging the urgency of real-time decision-making in AVs, the inclusion of detection time as a critical evaluation metric is made. This addition is important for assessing the real-time applicability and effectiveness of detection methods in operational TSC scenarios.

Through rigorous testing against the specific challenges posed by traffic sign datasets, this study bridges a significant gap in the current research landscape and paves the way for groundbreaking advancements in the safety and reliability of autonomous driving systems. The methodology and findings of this study provide a fresh perspective in the field, shedding light on the distinct nature of adversarial perturbations in TSC and their implications for AV navigation.

B. Paper Organization

The remaining sections of this paper are organized as follows: Section II provides an overview of adversarial attacks and defense measures, covering both IC and TSC domains. Section III presents the detection methods employed in this study and outlines the evaluation criteria used for their assessment. Section IV describes the experimental setup utilized, including the datasets and model configurations. Section V presents the results of our experiments, highlighting the performance of each evaluated detection method across various attacks and datasets. Finally, in Section VI, we conclude the paper by summarizing our findings and discussing their implications in TSC. We also highlight potential areas for future research and development, focusing on key aspects that require further investigation and improvement.

II. Related Works and Background Information

This section provides a comprehensive overview of the existing research and background information relevant to

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adversarial attacks and defense measures in both IC and TSC domains. It aims to establish the foundation for understanding the unique challenges and requirements of TSC in the context of adversarial threats. The subsections will explore the nature of adversarial attacks, various defense strategies employed in IC and TSC, and the current state of research in these areas.

A. Adversarial Attacks (AAs)

Most research on adversarial attacks has focused on well-known image classification datasets such as MNIST [33], CIFAR-10 [34], and ImageNet [35]. These attacks are typically classified based on the attacker's familiarity with the target model i.e. white-box attacks, where attackers have full access to the model, encompassing its architecture and parameters as well as its defense mechanism. Notable attack techniques include the Fast Gradient Sign Method (FGSM) [36], Basic Iterative Method (BIM) [37], Projected Gradient Descent (PGD) [38], DeepFool (DF) [39], and Carlini & Wagner (CW) [40] attacks. Each attack is uniquely designed to exploit the model's vulnerabilities.

1) FGSM [36] is an untargeted L_∞ attack that employs gradient to craft AEs. FGSM pinpoints the optimal perturbation direction for each pixel in an input image. This direction is derived from the sign of the gradient, with the intent to maximize the model's loss. The perturbation is then applied to the image in a single iteration, resulting in an AE. The mathematical representation of the FGSM attack is:

$$x' = x + \epsilon \cdot \text{sign}(\nabla_x L(\theta, x, y)) \tag{1}$$

where x is the input image, x' is the adversarial image, ϵ is the perturbation magnitude, ∇_x is the gradient with respect to x, $L(\cdot)$ is the loss function, θ are the parameters of the model, and y is the true label.

2) BIM [37], sometimes also referred to as the Iterative Fast Gradient Sign Method (I-FGSM), is an extension of the FGSM attack. While FGSM is a one-step attack, BIM iteratively applies the FGSM with a small step size, making it more effective in finding AEs. The adversarial perturbation is applied iteratively in small steps, ensuring that the perturbation remains within a specified limit. Mathematically, BIM can be described as:

$$\begin{cases} x'_0 = x \\ x'_{t+1} = \text{Clip}_{x,\epsilon}\left\{x'_t + \alpha \cdot \text{sign}(\nabla_x L(\theta, x'_t, y))\right\} \end{cases} \tag{2}$$

where α is the step size, and $\text{Clip}_{x,\epsilon}$ is the clipping function.

3) PGD [38] is an untargeted attack that refines AEs over multiple steps, similar to BIM. However, PGD introduces a projection step to ensure that the perturbations remain within a predefined specified ϵ-bound of the original input. Mathematically, PGD attack can be described as:

$$x'_{t+1} = \Pi_{\mathcal{X}}\left(x'_t + \alpha \cdot \text{sign}(\nabla_x L(\theta, x'_t, y))\right) \tag{3}$$

where $\Pi_{\mathcal{X}}$ is the projection function.

4) DF [39] is an iterative attack under the L_2 norm that finds the minimum amount of perturbation required to cause misclassification by measuring the distance from the original input to the decision boundary. Mathematically, DF attack can be written as:

$$x'_{t+1} = x'_t + \Delta x'_t \tag{4}$$

where $\Delta x'_t$ represents the minimal perturbation at iteration t, computed based on the linear approximation of the classifier's decision boundary.

5) CW [40] is an optimization-based attack targeting L_2, L_0, and L_∞ norms. Its objective is to find the minimum adversarial perturbation while considering a differentiable objective function that encourages misclassification and minimizes the perturbation magnitude. The CW attack can be described as the following optimization problem:

$$\begin{aligned} \min_{\Delta x} \quad & c \cdot \|\Delta x\|_p + f(x + \Delta x) \\ \text{subject to} \quad & x + \Delta x \in [0, 1]^n \end{aligned} \tag{5}$$

where Δx is the perturbation added, c is a constant that controls the trade-off between perturbation magnitude and classification confidence, f is the objective function designed to mislead the classifier, and n is the dimension of the input space.

It is worth noting that the CW attack was the first to successfully bypass defensive distillation [41], which was one of the most prominent defense mechanisms at the time.

In the context of TSC, adversarial attacks have received less attention compared to other IC domains. The unique characteristics of traffic sign images, such as their specific shapes, colors, and symbols, pose distinct challenges for designing effective adversarial attacks. Additionally, the consequences of misclassification in TSC can have significant safety implications. Therefore, understanding and mitigating adversarial attacks in the context of TSC is of utmost importance for ensuring the reliability and safety of AVs.

B. Defense Methods in IC

Defenses against AAs are generally divided into two main categories [42]: robustness-based and detection-based strategies. Robustness-based focus on strengthening the DNN models to ensure they can accurately classify images, even when they are subjected to adversarial manipulations. On the other hand, detection-based concentrates on identifying adversarial inputs at an early stage. Detection methods rigorously examine each image to determine whether it is normal or adversarial. Once an adversarial image is detected, it is immediately excluded from the classification process [24], [26], [28], thus preventing any potential hazard impact on the DNN's decision-making. In this study, the specific focus lies on detection methods as a defense against AAs.

Detection-based defenses against AAs are categorized into three main types: statistical methods, auxiliary models, and prediction inconsistency [28], [43], [44]. These categories offer varied strategies for identifying adversarial threats.

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Statistical methods utilize the inherent statistical properties of images and DNN layers. For instance, Kherchouche et al. [24] NSS method employs parameters from the Asymmetric Generalized Gaussian Distribution to detect changes in images caused by adversarial perturbations. Similarly, Ma et al. [25] use Local Intrinsic Dimensionality (LID) to measure density changes around data points, identifying areas where adversarial perturbations have altered the data's natural distribution. Feinman et al. [45] adopt Bayesian uncertainty estimates and deep feature subspace density estimation for detecting AEs.

Auxiliary models employ an additional model, known as the auxiliary model to differentiate between genuine and adversarial images before they are processed by the primary classification model. Aldahdooh et al. [26] developed the Selective and Feature-based Adversarial Detection (SFAD) method, leveraging SelectiveNet [46] to assess model uncertainty and predict confidence. SFAD examines intermediate feature maps, deriving new confidence probabilities to effectively identify AEs. Sotgiu et al. [27] DNR method employs Support Vector Machines (SVMs) to analyze feature anomalies, rejecting suspicious samples.

The prediction inconsistency approach relies on detecting inconsistencies in the predictions made by the model when presented with slightly altered versions of the same input. Xu et al. [28] FS method, employing bit-depth reduction and spatial smoothing, detects adversarial attacks by comparing predictions for original and altered inputs, revealing adversarial manipulations. Liang et al. [29] ANR method treats adversarial perturbations as noise, using uniform quantization and smoothing filters for mitigation. It compares classifications between original and noise-reduced images, efficiently spotting AEs without prior attack knowledge.

These methods have demonstrated high detection rates (DR) and low false positive rates (FPR) in IC. However, their effectiveness and potential limitations in TSC, a domain with its unique challenges, have yet to be thoroughly explored. Additionally, the detection time (DT) of these methods has not been extensively studied, especially in the context of TSC Therefore, there is a need to further investigate and study the detection time of these methods in the specific context of TSC. By considering the detection time, a better understanding of the practical feasibility and real-time applicability of these detection-based defense strategies in AV systems can be gained.

C. Defense Methods in TSC

There has been a notable increase in the development of defense methods against AAs, particularly in the form of robustness-based approaches. For example, Wu et al. [47] harnessed the power of singular value decomposition (SVD) combined with 5G and mobile edge computing to filter out adversarial components, ensuring real-time traffic sign classification. Li et al. [48] employed an attentional mechanism paired with affine coordinate mapping to redraw key traffic sign regions, mimicking human vision and effectively defending against multiple adversarial techniques. Additionally, Khan et al. [49] introduced a hybrid defense approach that combines random filtering, ensembling, and local feature mapping. This comprehensive approach effectively maintains high accuracy in traffic sign classification despite diverse adversarial attacks, thereby enhancing the model's overall robustness. These approaches aim to improve the resilience of models against adversarial attacks in TSC.

However, robustness-based methods in TSC face challenges. They often require significant modifications to the model architecture and retraining with adversarial samples, resulting in increased computational costs and the potential introduction of vulnerabilities to novel and unseen attack techniques [50]. Consequently, there is a growing trend towards adversarial detection-based defenses [42]. Detection-based methods offer a more cost-effective and efficient solution for protecting TSC systems against adversarial threats. Therefore, there is a need to shift the focus of defense strategies in the TSC domain towards detection-based approaches.

By exploring and developing detection-based defenses, researchers can mitigate the limitations associated with robustness-based methods while still ensuring the integrity and reliability of TSC systems for AVs. Detection-based approaches provide an opportunity to identify and exclude adversarial inputs in real-time, without extensive model modifications or retraining. This shift in focus towards detection-based defenses represents a promising direction for the advancement of robust and efficient security measures in TSC. An illustrative example is the Real-time Cloud-based Adversarial-Intrusion Detection System (RCA-IDS) [51], which employs real-time cloud-based IDS to detect and mitigate attacks in connected and autonomous vehicle systems. By leveraging cloud-based processing and machine learning, RCA-IDS demonstrates the potential to enhance the robustness and scalability of detection methods, ensuring timely and accurate identification of adversarial threats in TSC.

III. Detection Methods and Their Evaluation Criteria

This section discusses adapting six detection methods from the IC domain to TSC. These methods have been selected based on their effectiveness in detecting adversarial attacks and are evaluated under uniform conditions across three TSC datasets, models, and attack scenarios. Each subsection provides a detailed technical analysis of the methods, their configurations, and hyperparameters. The evaluation criteria used to assess the effectiveness and efficiency of these methods are also described. To provide a clearer comparison, we have included a pros/cons table (Table III) summarizing the strengths, weaknesses, and performance results of each detection method. Additionally, we have included detailed pseudocode algorithms in Table V and computational complexity descriptions in Table III for each of the six methods.

A. Detection Methods

1) Natural Scene Statistical (NSS) Method Previous research has shown that the statistical properties of adversarial examples (AEs) differ from clean images [24], [25]. Kherchouche et al. [24] used the Asymmetric

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TABLE I
CLASSIFIER'S LAYERS UTILIZED IN SFAD [26] METHOD

Classifier's Layers	TSRD	BTSC	GTSRB
1st Classifier	Dropout	Dense + ReLU	Dense + ReLU
2nd Classifier	Dense + ReLU	Dense + ReLU	Dense + ReLU

Generalized Gaussian Distribution (AGGD) from the Mean Subtracted Contrast Normalized (MSCN) coefficients to train a binary SVM classifier, extracting 18 features from each grayscale image.

We adapted this approach by considering each channel of an RGB image separately, extracting 54 features per image to train a binary SVM classifier. This adaptation leverages the rich color information crucial for accurate traffic sign identification, enhancing the NSS method's accuracy in detecting adversarial modifications. The PGD attack was used during training as a representative due to its proven generalizability across various attacks [38].

2) Local Intrinsic Dimensionality (LID) Method The LID method detects AEs by analyzing the local dimensional properties around data points [25]. The key steps of LID are:
   a) Noisy sample generation: To handle noisy samples as if they were clean samples, Gaussian noise is added to each pixel. The scale of the added noise is determined based on the Lp-norm of the corresponding adversarial perturbation.
   b) LID Score Calculation: The LID score is computed for each sample based on its k-nearest neighbor samples.
   c) Classification: A logistic regression classifier is trained using the LID scores obtained from both the noisy and clean samples. The aim is to distinguish between AEs and clean samples based on their LID scores.

   For TSC, we set the value of k for LID estimation to 10 for all TSRD, GTSRB, and BTSC datasets, instead of the values used in the original work with MNIST and CIFAR-10 [25]. This adaptation, along with using a logistic regression classifier, enables effective AE detection in TSC.
3) Deep Neural Reduction (DNR) Method The DNR method employs a multi-classifier system based on Radial Basis Function (RBF) SVMs to identify and reject samples with unusual features [27]. It involves training three RBF-SVM classifiers on distinct feature maps from the last three layers of the network, with a final RBF-SVM trained on the prediction probabilities of the first three SVMs. Only clean datasets are used during training. For testing, a 10% threshold from the maximum prediction probabilities of clean samples is used to differentiate between AEs and clean images.

   For TSC, we streamlined the process by using two classifiers for all TSRD, BTSC, and GTSRB datasets, as illustrated in Figure 1. Originally, the DNR method utilized three RBF-SVM classifiers on feature maps from the last three layers of the network. Our adaptation uses two classifiers and incorporates the Dropout layer, replacing the three layers used in the original method. These modifications improve time efficiency and reduce computational complexity. Additionally, the RBF kernels in the SVMs were replaced with linear kernels, enhancing classification results for TSC. The predefined threshold was adjusted to 20% to improve detection accuracy for traffic sign scenarios.

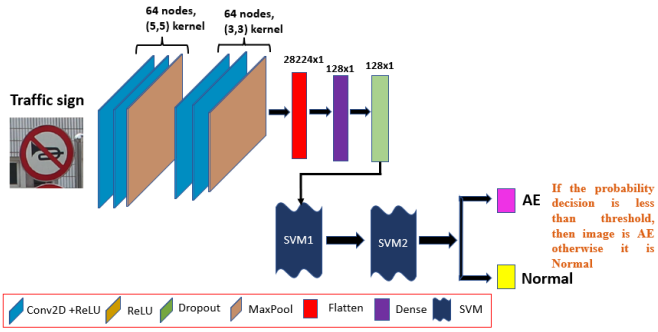


Fig. 1. Adapted DNR method framework for TSC.

4) Selective and Feature-based Adversarial Detection (SFAD) Method The SFAD method uses SelectiveNet to build a detector [46]. Three classifiers are trained using SelectiveNet, each fed with feature maps from specific layers of a baseline CNN classifier. These feature maps undergo various processing steps, including autoencoding, up/down sampling, bottleneck, and noise blocks. A fourth classifier is trained using the prediction probabilities from the previous classifiers.

   In our TSC adaptation, SFAD utilizes two classifiers as shown in Table I to reduce computation complexity. The three approaches used (shown in Figure 2) are:
   a) Approach F: A normal DNN classifier processes traffic images into their respective classes, with feature maps from the last two layers used as input in the second stage
   b) Approach M: Two blocks, each with a Processing Layer, Dense Layer, and SelectiveNet, process individual feature maps from Approach F.
   c) Approach S: A concatenation layer followed by a Dense Layer and SelectiveNet, which receives input probabilities from Approach M, and outputs the argmax of the probabilities.

   During training, threshold values are determined to classify the argmax as adversarial or non-adversarial. For TSRD, BTSC, and GTSRB datasets, the coverage and coverage threshold values for the first and second classifiers are (0.9, 0.9), while for the third classifier, it is (0.9, 0.7). These threshold values yielded the best results in our experiments.
5) Feature Squeezing (FS) Method The FS method detects AEs by transforming input images through various 'squeezers' and analyzing changes in prediction probabilities [28]. Specific squeezers are tailored for each dataset to impact adversarial perturbations more significantly than legitimate image content.

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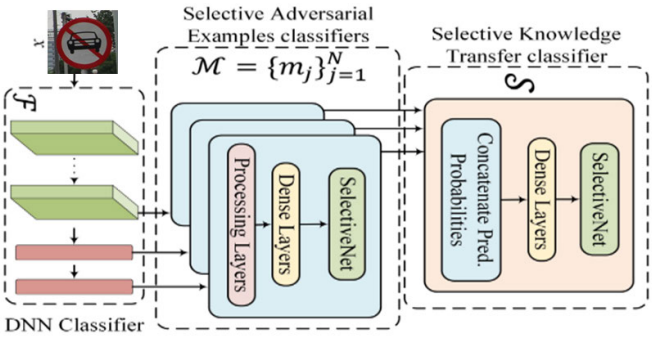


Fig. 2. The three approaches used in SFAD method [26].

In our TSC adaptation, we applied a universal set of squeezers for all TSRD, GTSRB, and BTSC datasets: 5-bit color depth reduction, 2×2 median smoothing, and an 11-3-2 non-local mean denoiser filter. FS calculates the maximum L_1 or L_2 distance between the prediction probabilities of original and squeezed samples. A threshold value of 5% is set to distinguish between normal and adversarial examples.

6) Adaptive Neural Reduction (ANR) Method The ANR method detects AEs by treating adversarial perturbations as noise and uses adaptive image processing techniques, such as scalar quantization and smoothing spatial filters, to eliminate adversarial perturbations based on image entropy [29]. Key approaches include:
   a) AE Generation for Training: Utilizing FGSM to generate AEs from clean images during the training phase.
   b) Preservation of Original Pixel Levels: IMaintaining the original pixel levels of the images (ranging from [0, 255]) without normalization, allowing proper calculation of entropy and quantization.
   c) Scalar Quantization Approach: Applying scalar quantization based on image entropy levels.
   d) Spatial Smoothing Filter Approach: Applying a spatial smoothing filter based on entropy levels.

   For TSC, we adapted ANR by:
   a) Computation of Entropy: Calculating the entropy of each image to determine the appropriate quantization level.
   b) Scalar Quantization Approach: Applying quantization with interval sizes of 64 (entropy < 4.0), 32 (entropy between 4 and 5), and 16 (entropy > 5.0). After quantization, images are either classified directly or proceed to the smoothing process.
   c) Spatial Smoothing Filter Approach: Applying a mean filter of size 8 to images with (entropy $>$ 5.0).

B. Evaluation Metrics

1) Detection Rate (DR): The detection rate, also known as the true positive rate, measures the detector's accuracy in identifying successful AEs. It is calculated by counting the number of AEs that are accurately predicted by the detector and dividing it by the total number of AEs. A higher detection rate indicates a better performance of the detector in effectively recognizing and classifying adversarial examples within the TSC.
2) False Positive Rate (FPR): measures the proportion of clean inputs that are mistakenly classified as adversarial inputs by the detector. FPR is calculated by dividing the number of falsely detected clean inputs as adversarial by the total number of clean inputs. A lower FPR indicates a better performance of the detector in accurately identifying and preserving the integrity of clean inputs, without misclassifying them as adversarial inputs.
3) Detection Time (DT): It is the time taken by the detector to determine whether an input is adversarial or not. It represents the speed at which the detector processes and analyzes the input data. In TSC applications, minimizing detection time is crucial for real-time decision-making and ensuring the efficiency of autonomous vehicles. A shorter detection time enables faster responses and enhances the overall performance of TSC systems, contributing to safer and more reliable navigation.

In the evaluation process, the initial stages of capturing and preprocessing traffic sign images are consistent across all methods. However, the subsequent feature extraction and adversarial identification stages may differ among the methods, as each employs its unique techniques. To ensure a fair comparison, the evaluation takes into account the varying capabilities of different attack models, to generate the same number of successful adversarial samples. The detection time is then calculated as the average time spent on feature extraction and detection for one adversarial image. By considering these evaluation metrics, the effectiveness and efficiency of detection methods can be assessed and compared, facilitating the development of robust defense mechanisms against adversarial attacks in TSC.

IV. Exrerimental Setup

This section describes the experimental setup used to evaluate the detection methods. It includes details on the Hardware setup, datasets, target classifiers, and attack scenarios. It explains how the experiments were conducted to ensure a fair comparison of the methods' performance in detecting adversarial examples in TSC.

A. Hardware Setup

The experiments in this study were conducted using a workstation with the following hardware setup: an Intel (R) Core (TM) i7-7700 CPU running at a speed of 3.60 GHz, 32 GB of RAM, and an NVIDIA GeForce RTX 4080 GPU with 12 GB of RAM. The workstation was equipped with Windows 11 (64-bit) as the operating system. For the implementation of the networks, Python 3.9 was used as the programming language, with the Keras framework and TensorFlow 2.5 as the backend for deep learning computations.

B. Datasets

Three distinct datasets were utilized in this study: TSRD, BTSC, and GTSRB, as shown in Table II. These datasets

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TABLE II
DATASET'S AND CLASSIFIER'S INFORMATION

No	Datasets	Country	Classes	Training Samples	Testing Samples	Classifiers	Classifier's Accuracy
1	TSRD	China	58	4170	1994	CNN*	0.9935
2	GTSRB	Germany	43	39209	12630	VGG16 [52]	0.8538
3	BTSC	Belgium	62	4591	2534	InceptionV3 [53]	0.9692

*The CNN classifier is developed as part of this work.

TABLE III
SUMMARY OF DETECTION METHODS

Method	Description	Pros	Cons	Performance Summary			
				DR	FPR	DT	CC
NSS [24]	Uses MSCN coefficients to detect statistical anomalies in image properties	Effective across all attacks	Needs AEs during training	*****	*****	*****	*****
LID [25]	Detects AEs by measuring changes in local dimensionality	Effective in high-dimensional data	Computationally expensive	****	*****	**	****
SFAD [26]	Estimates uncertainties in DNN last layers outputs using SelectiveNet	Does not require prior knowledge of attacks	Requires tuning for optimal performance	***	***	******	**
DNR [27]	Rejects inputs with abnormal feature patterns across layers	No need for AEs during training	Requires multiple network layers and end-to-end training for best performance	*	**	***	***
FS [28]	Detects AEs by simplifying inputs through bit depth reduction and smoothing, then comparing predictions	Simple and inexpensive	May not handle larger perturbations well	**	****	****	****
ANR [29]	Reduces noise with entropy-scalar quantization and smoothing, then compares classification results	Most effective across all attacks	May degrade clean image quality	******	*	*	******

"*": This indicates the lowest performance in the evaluation metric.
"****"**: This indicates the highest performance in the evaluation metric.
"CC": This is the computational complexity mainly time and space have been considered.

were selected based on their alignment with international road sign standards and their recognition within the field of TSC research. They provide a diverse range of traffic sign designs and encompass various real-world conditions, including different weather and lighting scenarios, as well as varying levels of occlusions. This diversity is crucial for thoroughly evaluating the effectiveness of detection methods across different global contexts. By incorporating these datasets, the study adheres to research authorization standards and ensures robust findings that are comprehensive and applicable on a global scale. This approach acknowledges the diverse nature of TSC challenges encountered in autonomous driving systems worldwide.

To optimize the datasets for training and testing classifiers, preprocessing was performed. This involved cropping and resizing the images to a standardized size of 100×100 pixels. Additionally, the pixel values were normalized to the range of [0, 1]. As part of the preprocessing, the BTSC dataset was modified to include 32 out of the original 62 classes. This modification allowed for more effective training and testing of the classifiers.

C. Target Classifiers

In the experimental design, three distinct classifiers were utilized, each tailored to a specific dataset, to demonstrate the model-agnostic nature of the detection methods. These classifiers were chosen to represent a range of architectures commonly employed in TSC tasks. For the TSRD dataset, a 10-layer CNN Classifier was developed. The architecture of this classifier is outlined in Table IV. For the BTSC and GTSRB datasets, pre-trained VGG16 [52] and Inception V3 [53] classifiers were used, respectively. The accuracies of these classifiers can be seen in Table II.

All classifiers were trained using the rectified linear unit (ReLU) activation function and cross-entropy loss. The models were optimized using the Adam optimizer with a learning rate of 0.001. A uniform batch size of 32 and 30 epochs were employed for training across all datasets. By employing this diverse set of classifiers, the study aimed to demonstrate that the effectiveness of the detection methods extends beyond the specific choice of classifier. This highlights the universal applicability of the methods in the field of TSC.

D. Implementation of Attacks

The implementation of adversarial attacks in this study utilized version 3.3.3 of the Foolbox library.[1]

To assess the effectiveness of the detection methods, a range of state-of-the-art adversarial attacks was generated. The focus

[1]https://foolbox.readthedocs.io/en/stable/index.html

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TABLE IV
TSRD CLASSIFIER'S ARCHITECTURE

No	Layer	Description
1	Conv2D + ReLU	64 filters (5x5)
2	Conv2D + ReLU	64 filters (5x5)
3	MaxPooling2D	(2x2)
4	Conv2D	64 filters (3x3)
5	Conv2D + ReLU	64 filters (3x3)
6	MaxPooling2D	(2x2)
7	Flatten	
8	Dense + ReLU	128 units
9	Dropout	0.1
10	Dense + SoftMax	58 units

was on different L_p -norm attacks, covering various scenarios. Specifically, the following attack categories were utilized:

1) L_2 Norm Attacks: The employed attacks in this category are CW and DF attacks.
2) L_∞ Norm Attacks: This category included the FGSM, BIM, and PGD attacks.

For the L_∞ norm attacks, a range of ϵ values was selected to comprehensively evaluate the detector's performance. The chosen values were 0.05, 0.1, 0.15, 0.2, 0.25, and 0.3. These values struck a balance between generating effective adversarial examples and avoiding perceptible perturbations that could be easily detected by humans.

Regarding the CW attack, extensive testing was conducted with 1000 iterations. The confidence parameter κ was varied in the range 0, 0.5, 1, and 2 to comprehensively assess the detection methods under diverse adversarial conditions. These parameter ranges are commonly employed in similar studies and ensure a practical and realistic evaluation of the detector's robustness [29], [54].

E. Implementation of Detection Methods

The implemented detection methods in this study followed the versions provided by the original authors. This approach ensured consistency with the specifications and parameters outlined in the respective research papers. By adopting this approach, the performance of the detection methods for TSC in the context of AVs could be accurately evaluated and compared.

V. RESULTS AND DISCUSSION

This section evaluates the performance of six detection methods on three datasets against five white-box attacks. The evaluation includes the Detection Rate (DR), False Positive Rate (FPR), and Detection Time (DT) of the tested detectors on the TSRD, BTSC, and GTSRB datasets, as presented in Table VI, Table VII, and Table VIII, respectively. The performance comparison of the detection methods against white-box attacks based on the detection rate and time is summarized in Figure 3 and Figure 4, respectively.

This evaluation forms the basis of our comparative study, enabling the assessment of the adaptability and effectiveness of these detection techniques in TSC. Through a thorough examination of the performance results, valuable insights are provided to enhance the safety and reliability of AVs by improving the robustness of TSC systems against adversarial threats. Based on the results displayed in Tables VI, VII, VIII, and Figures 3 and 4, the observations on each method are given in the following subsections (V-A to V-F).

A. Natural Scene Statistical (NSS) Method

The NSS method demonstrates strong performance in terms of detection rate across different datasets and attacks. Specifically, on the GTSRB dataset, NSS performs exceptionally well against all attacks. On the BTSC dataset, it achieves great performance, except for CW attacks where the performance is medium. Similarly, on the TSRD dataset, NSS shows excellent performance, with the exception of DF and CW attacks.

It is worth noting that NSS exhibits remarkable performance in detecting PGD attacks across all datasets, as it was trained solely with PGD-based AEs. The features learned from PGD generalize well to other attacks, as evidenced by the effectiveness of NSS against FGSM and BIM attacks across all datasets. This suggests that incorporating other AEs during the training process could further enhance the detector's performance. Additionally, the NSS detector demonstrates very low FPR across all datasets, indicating its reliability.

In terms of detection time, the NSS method exhibits impressively low detection times across all datasets. For the TSRD dataset, the detection time is 0.0243 seconds, while for the BTSC dataset, it is 0.0145 seconds. The GTSRB dataset shows a slightly higher detection time of 0.029 seconds. This efficiency is achieved by leveraging the parallel processing capability of the feature extraction process alongside the prediction process of the baseline classifier.

Overall, the NSS method showcases strong detection rates and low detection times, making it a promising approach for adversarial example detection across various datasets and attack scenarios on TSC.

B. Local Intrinsic Dimensionality (LID) Method

The LID method exhibits varying performance across different datasets and attacks. On the TSRD dataset, it performs well against most attacks, except for a medium performance on the DF attack. Conversely, on the BTSC and GTSRB datasets, it demonstrates better performance against all attacks.

Interestingly, the detection rate of the LID method tends to increase with the level of perturbation for gradient-based attacks (FGSM, BIM, PGD) on both the TSRD and GTSRB datasets. This suggests that as the level of perturbation increases, the LID method becomes more effective in distinguishing between clean and adversarial images. The changes in the local neighborhood structure become more pronounced with larger perturbations, making it easier for LID to detect the presence of adversarial examples.

Regarding FPR, the LID demonstrates low FPR on both the BTSC and GTSRB datasets. However, it exhibits a higher FPR on the TSRD dataset, particularly for CW attacks, which are known to be among the strongest attacks.

Regarding detection time, the LID method proves to be the second most time consuming of all the methods in all databases. The reason behind the high inference time observed

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TABLE V
PSEUDOCODE FOR THE SIX METHODS ALGORITHMS

Input: Image I, Output: Adversarial/Non-Adversarial

NSS-Based Detection

```
Function NSS_Detection(I):
    RGB_I = SplitChannels(I)
    for (each channel in RGB_I):
        Channel_Features = ExtractMSCNFeatures(channel)
        RGB_Features = Concatenate(Channel_Features)
    End for
    Train SVM (RGB_Features from PGD)
    Test SVM (all attacks)
End Function
```

LID-Based Detection

```
Function LID_Detection(I):
    # Training Phase
        Noisy_I(NI) = Add Gaussian Noise(I)
        LID_Scores_I = ComputeLIDScores(I)
        LID_Scores_NI = ComputeLIDScores(NI)
        Train Logistic Regression (LID_Scores_I, LID_Scores_I(NI))
    # Testing Phase
        if (Score == LID_score_I):
            I = Non-Adversarial
        else:
            I = Adversarial
        End if
End Function
```

DNR-Based Detection

```
Function DNR_Detection(I):
    Features = ExtractFeatures(I)
    # Training Phase(clean images only)
        Prediction1 = SVM1(Features)
        Prediction2 = SVM2(Prediction1)
    # Testing Phase(clean and adversarial images)
        Features = ExtractFeatures(All images)
        Prediction1 = SVM1(Features)
        Prediction2 = SVM2(Prediction1)
        if (Prediction2 =< Threshold):
            I = Adversarial
        else:
            I = Non-Adversarial
        End if
End Function
```

SFAD-Based Detection

```
Function SFAD_Detection(I):
    Deep_Features = Two_Last_Layers_CNN(I)
    # Training Phase(clean images only)
        Prediction1 = Two_Selective_Classifiers(Deep_Features)
        Prediction2 = Selective_Knowledge_Classifier(Prediction1)
    # Testing Phase(clean and adversarial images)
        Deep_Features = Two_Last_Layers_CNN(I)
        Prediction1 = Two_Selective_Classifiers(Deep_Features)
        Prediction2 = Selective_Knowledge_Classifier(Prediction1)
        if (Predictions =< Threshold):
            I = Adversarial
        else:
            I = Non-Adversarial
        End if
End Function
```

FS-Based Detection

```
Function FS_Detection(I):
    Squeezed_I = ApplySqueezers(I)
    Original_Prediction = Model_Predict(I)
    Squeezed_Prediction = Model_Predict(Squeezed_I)
    Distance = Distance(Original_Prediction, Squeezed_Prediction)
    if (Distance > Threshold):
        I = Adversarial
    else:
        I = Non-Adversarial
    End if
End Function
```

ANR-Based Detection

```
Function ANR_Detection(I):
    Entropy = CalculateEntropy(I)
    Quantized_I = ApplyQuantization(I, Entropy)
    if (Entropy > Threshold):
        Smoothed_I = ApplySmoothing(Quantized_I)
    else:
        Smoothed_I = Quantized_I
    End if
    Original_Prediction = Model_Predict(I)
    Smoothed_Prediction = Model_Predict(Smoothed_I)
    if (Original_Prediction != Smoothed_Prediction):
        I = Adversarial
    else:
        I = Non-Adversarial
    End if
End Function
```

in LID could be due to the involvement of Euclidean distance (L2 distance) calculations between the reference points and other points within a specified neighborhood range when estimating the local intrinsic dimensionality [25]. Additionally, the detection time is prolonged as the method processes and analyzes a larger number of features in each image making it unsuitable for AVs

C. Selective and Feature-Based Adversarial Detection (SFAD) Method

The SFAD method demonstrates consistently strong performance across various attacks and datasets. Specifically, on the TSRD dataset, SFAD performs exceptionally well against all attacks, particularly CW attacks. On the BTSC and GTSRB datasets, SFAD exhibits excellent performance against FGSM, DF, and CW attacks, while achieving medium performance against BIM and PGD attacks. The success of SFAD can be attributed to its ensemble nature, which combines uncertainty methods through selective rejection, confidence probability, and bi-modal mismatch detection. SFAD showcases impressive performance in terms of minimizing false detections, as indicated by its low FPR across all datasets.

In addition to its strong performance, SFAD stands out as the most efficient method among all the detection methods evaluated on the datasets. It boasts remarkably low detection times, making it highly suitable for autonomous vehicles. For the TSRD dataset, the detection time is as lower as 0.0077 seconds, while for the BTSC dataset, it is even lower at 0.0062 seconds. The GTSRB dataset shows a slightly higher detection time of 0.0088 seconds, which remains quite efficient.

Overall, the results highlight SFAD's robust performance across attacks and datasets, along with its ability to minimize false detections. Moreover, its exceptional efficiency in terms

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TABLE VI

DETECTION RATE (DR), FALSE POSITIVE RATE (FPR) AND DETECTION TIME (DT) FOR THE SIX DETECTION METHODS ON TSRD DATASET. THE HIGHEST DETECTION RATE AND THE LOWEST DETECTION TIME ARE HIGHLIGHTED IN **BOLD** FOR EACH METHOD

TSRD Dataset																			
Detection Methods Categories																			
Attacks		**Statistical Method**						**Auxiliary Model**						**Prediction Inconsistency**					
		NSS			**LID**			**DNR**			**SFAD**			**FS**			**ANR**		
		DR	**FPR**	**DT** (10^{-2})	**DR**	**FPR**	**DT** (10^{-2})	**DR**	**FPR**	**DT** (10^{-2})	**DR**	**FPR**	**DT** (10^{-2})	**DR**	**FPR**	**DT** (10^{-2})	**DR**	**FPR**	**DT** (10^{-2})
Method	**Parameter**	**(%)**	**(%)**	**(sec)**	**(%)**	**(%)**	**(sec)**	**(%)**	**(%)**	**(sec)**	**(%)**	**(%)**	**(sec)**	**(%)**	**(%)**	**(sec)**	**(%)**	**(%)**	**(sec)**
FGSM	ϵ = 0.05	98.50	0.08	2.97	50.24	1.90	15.42	37.54	20.00	11.83	61.08	6.50	**0.20**	**65.38**	5.88	2.04	98.00	3.00	47.33
FGSM	ϵ =0.1	97.71	0.08	1.64	71.64	6.71	15.36	34.84	20.00	12.16	73.08	5.92	0.21	60.77	5.38	1.96	98.00	4.00	45.80
FGSM	ϵ =0.15	96.42	0.08	1.63	82.06	5.21	15.51	36.39	20.00	9.91	77.00	5.25	**0.20**	58.18	5.43	1.97	95.00	1.00	51.00
FGSM	ϵ =0.2	92.38	0.08	1.58	87.21	3.49	15.67	36.48	20.00	12.59	77.67	5.42	0.22	57.58	4.71	1.94	**99.00**	2.00	47.57
FGSM	ϵ =0.25	83.96	0.08	1.58	89.17	3.61	16.47	37.13	20.00	11.92	78.46	5.83	2.20	56.71	5.96	1.96	**99.00**	3.00	46.98
FGSM	ϵ =0.3	72.62	0.08	3.28	**93.13**	1.88	17.54	39.59	20.00	12.66	76.13	6.33	2.20	56.78	5.85	1.93	**99.00**	2.00	48.72
BIM	ϵ =0.05	59.75	0.08	3.24	55.61	8.77	16.44	43.52	20.00	10.82	71.92	5.42	0.23	54.94	5.62	1.97	95.00	1.00	47.02
BIM	ϵ =0.1	84.17	0.08	3.26	69.41	11.7	17.96	57.79	20.00	10.67	72.71	6.42	0.21	46.97	4.59	1.97	95.00	1.00	44.80
BIM	ϵ =0.15	93.88	0.08	3.30	76.42	10.5	17.51	66.48	20.00	7.48	75.58	5.67	0.22	47.37	5.25	2.05	96.00	2.00	48.10
BIM	ϵ =0.2	96.33	0.08	3.30	78.34	8.82	**12.70**	68.44	20.00	9.45	72.46	5.33	0.22	47.70	4.59	1.93	98.00	2.00	48.12
BIM	ϵ =0.25	97.38	0.08	1.43	81.66	8.44	12.94	74.59	20.00	8.09	78.46	6.08	0.22	45.66	5.78	1.96	97.00	4.00	45.35
BIM	ϵ =0.3	97.88	0.08	1.70	84.63	7.81	12.73	69.18	20.00	8.98	76.52	6.33	0.23	47.79	4.43	1.95	95.00	6.00	45.42
PGD	ϵ =0.05	98.79	0.08	3.26	63.39	12.20	17.60	55.98	20.00	11.38	66.21	6.42	0.22	51.70	5.20	1.95	95.00	3.00	45.81
PGD	ϵ =0.1	99.79	0.08	1.71	76.28	10.78	17.79	68.61	20.00	10.20	73.29	6.08	0.21	46.97	4.59	1.95	95.00	0.00	46.04
PGD	ϵ =0.15	**99.92**	0.08	1.75	79.60	9.33	18.07	70.98	20.00	7.71	73.50	5.75	0.22	47.46	5.25	2.05	97.00	1.00	44.77
PGD	ϵ = 0.2	**99.92**	0.08	1.64	80.64	8.75	17.43	68.28	20.00	8.70	72.21	6.33	0.22	47.70	4.59	2.00	96.00	0.00	46.95
PGD	ϵ =0.25	99.88	0.08	1.71	80.16	8.47	17.52	69.84	20.00	7.87	77.63	5.58	0.22	45.66	5.67	1.93	95.00	2.00	48.59
PGD	ϵ =0.3	**99.92**	0.08	1.63	81.75	7.67	17.65	75.16	20.00	8.26	73.42	6.75	0.21	47.79	4.43	**1.92**	99.00	3.00	44.82
DF	-	49.44	0.08	2.93	50.00	0.00	17.81	46.39	20.00	12.62	84.63	6.33	2.20	62.47	5.83	1.95	98.00	1.00	39.73
CW	κ = 0.0	50.00	0.08	2.97	60.95	18.16	15.42	92.38	20.00	9.50	95.88	5.92	0.44	64.22	4.79	1.96	91.00	3.00	41.63
CW	κ = 0.5	50.02	0.08	2.90	61.32	17.41	15.44	95.91	20.00	9.09	95.01	6.58	0.27	66.22	3.11	1.94	88.00	1.00	41.50
CW	κ = 1.0	49.98	0.08	2.73	75.75	19.50	15.50	96.48	20.00	9.50	95.71	5.92	0.23	49.83	5.79	1.95	90.00	2.00	**38.99**
CW	κ = 2.0	50.00	0.08	**1.12**	79.00	18.00	16.77	**100.00**	20.00	**5.71**	96.54	6.92	0.32	46.93	6.15	1.99	95.00	2.00	41.23

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TABLE VII

DETECTION RATE (DR), FALSE POSITIVE RATE (FPR) AND DETECTION TIME (DT) FOR THE SIX DETECTION METHODS ON BTSC DATASET. THE HIGHEST DETECTION RATE AND THE LOWEST DETECTION TIME ARE HIGHLIGHTED IN **BOLD** FOR EACH METHOD

BTSC Dataset																			
Detection Methods Categories																			
Attack		**Statistical Method**						**Auxiliary Model**						**Prediction Inconsistency**					
		NSS			**LID**			**DNR**			**FSAD**			**FS**			**ANR**		
		DR	**FPR**	**DT** (10^{-2})	**DR**	**FPR**	**DT** (10^{-2})	**DR**	**FPR**	**DT** (10^{-2})	**DR**	**FPR**	**DT** (10^{-2})	**DR**	**FPR**	**DT** (10^{-2})	**DR**	**FPR**	**DT** (10^{-2})
Method	**Parameter**	**(%)**	**(%)**	**(sec)**	**(%)**	**(%)**	**(sec)**	**(%)**	**(%)**	**(sec)**	**(%)**	**(%)**	**(sec)**	**(%)**	**(%)**	**(sec)**	**(%)**	**(%)**	**(sec)**
FGSM	ϵ = 0.05	**99.95**	0.05	1.45	83.06	3.44	34.98	76.64	19.78	10.69	69.14	4.61	0.58	68.58	5.94	2.60	94.19	27.40	24.24
FGSM	ϵ =0.1	**99.95**	0.05	1.42	88.59	1.36	**30.32**	82.13	19.83	10.82	60.08	5.22	0.65	71.53	5.69	2.56	95.00	27.40	22.22
FGSM	ϵ =0.15	**99.95**	0.05	1.70	95.86	1.74	34.68	90.74	19.83	10.79	80.14	4.67	0.62	79.76	5.49	2.56	94.80	27.40	22.15
FGSM	ϵ =0.2	**99.95**	0.05	1.29	98.39	1.19	33.94	95.66	19.83	10.67	83.81	5.06	0.61	81.93	5.66	2.54	94.39	27.40	21.82
FGSM	ϵ =0.25	**99.95**	0.05	1.44	**99.01**	1.18	33.97	98.28	19.78	11.02	97.42	4.78	0.61	91.38	4.91	2.56	96.39	27.40	21.96
FGSM	ϵ =0.3	**99.95**	0.05	1.66	98.47	0.68	35.24	99.92	19.83	10.56	83.19	4.89	0.63	88.40	5.69	2.55	94.19	27.40	21.86
BIM	ϵ =0.05	54.87	0.05	1.41	85.00	1.41	35.15	1.83	19.78	8.36	47.69	4.61	0.63	47.66	5.11	2.62	95.00	27.40	21.96
BIM	ϵ =0.1	63.76	0.05	1.52	88.36	0.78	34.93	1.00	19.83	8.15	47.75	4.50	0.63	46.30	5.11	2.58	93.80	27.40	22.59
BIM	ϵ =0.15	82.47	0.05	1.44	92.50	0.94	35.11	2.06	19.83	8.19	48.08	4.33	0.63	47.45	5.11	2.55	95.39	27.40	**21.54**
BIM	ϵ =0.2	99.47	0.05	1.52	90.31	0.31	35.09	2.94	19.83	8.39	47.69	4.61	0.69	46.85	5.43	**2.52**	94.39	27.40	21.99
BIM	ϵ =0.25	99.89	0.05	1.73	93.21	0.36	34.83	3.67	19.78	8.09	47.58	4.83	0.67	47.72	4.57	2.59	95.00	27.40	21.78
BIM	ϵ =0.3	96.95	0.05	1.41	93.30	0.71	35.21	4.28	19.83	8.37	47.69	4.61	0.62	47.29	5.43	2.53	95.80	27.40	21.86
PGD	ϵ =0.05	**99.95**	0.05	1.41	82.86	0.36	32.67	0.50	19.78	8.40	47.53	4.94	0.65	47.45	5.11	2.57	98.50	20.00	65.20
PGD	ϵ =0.1	**99.95**	0.05	1.47	90.63	0.78	33.41	1.94	19.83	8.27	48.69	4.94	0.62	46.30	5.11	2.56	94.50	20.00	65.26
PGD	ϵ =0.15	**99.95**	0.05	1.46	85.27	0.18	33.12	2.94	19.83	8.23	51.92	4.72	0.67	47.45	5.11	2.54	95.50	20.00	69.17
PGD	ϵ = 0.2	**99.95**	0.05	1.49	89.30	0.63	33.41	2.83	19.83	8.45	49.17	4.78	0.62	46.85	5.43	2.54	95.50	20.00	68.19
PGD	ϵ =0.25	**99.95**	0.05	1.47	88.36	0.94	33.10	2.17	19.78	8.23	47.39	5.28	0.64	47.72	4.57	2.57	94.50	20.00	69.23
PGD	ϵ =0.3	**99.95**	0.05	1.55	92.03	0.63	33.68	2.11	19.83	8.43	48.31	4.61	0.66	47.28	5.43	2.54	94.00	20.00	66.66
DF	-	90.79	0.05	1.54	88.53	0.95	35.19	92.87	19.78	8.13	60.17	4.56	0.66	89.31	3.20	2.60	89.00	11.00	64.25
CW	κ = 0.0	50.00	0.05	1.26	83.04	5.00	35.20	98.20	19.78	8.12	78.47	5.00	0.60	**98.65**	2.39	2.64	98.00	11.00	59.85
CW	κ = 0.5	50.00	0.05	1.40	76.72	2.03	35.15	98.20	19.83	8.67	69.05	5.00	0.63	96.37	6.60	2.62	93.00	11.00	62.26
CW	κ = 1.0	50.00	0.05	**1.08**	82.49	1.01	32.03	99.51	19.83	8.13	**97.69**	4.61	**0.35**	47.37	5.26	**2.52**	96.00	11.00	63.36
CW	κ = 2.0	50.00	0.05	1.26	69.36	0.19	33.59	**100.00**	19.83	**5.91**	47.61	4.78	0.60	46.88	6.25	**2.52**	**100.00**	11.00	53.25

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TABLE VIII

DETECTION RATE (DR), FALSE POSITIVE RATE (FPR) AND DETECTION TIME (DT) FOR THE SIX DETECTION METHODS ON GTSRB DATASET. THE HIGHEST DETECTION RATE AND THE LOWEST DETECTION TIME ARE HIGHLIGHTED IN **BOLD** FOR EACH METHOD

BTSC Dataset																			
Detection Methods Categories																			
Attack		**Statistical Method**						**Auxiliary Model**						**Prediction Inconsistency**					
		NSS			**LID**			**DNR**			**FSAD**			**FS**			**ANR**		
		DR	**FPR**	**DT**	**DR**	**FPR**	**DT**	**DR**	**FPR**	**DT**	**DR**	**FPR**	**DT**	**DR**	**FPR**	**DT**	**DR**	**FPR**	**DT**
				(10^{-2})			(10^{-2})			(10^{-2})			(10^{-2})			(10^{-2})			(10^{-2})
Method	**Parameter**	**(%)**	**(%)**	**(sec)**	**(%)**	**(%)**	**(sec)**	**(%)**	**(%)**	**(sec)**	**(%)**	**(%)**	**(sec)**	**(%)**	**(%)**	**(sec)**	**(%)**	**(%)**	**(sec)**
FGSM	ϵ = 0.05	**99.96**	0.04	3.38	90.13	7.54	27.36	71.89	20.02	11.49	72.89	11.11	**0.47**	60.40	5.22	2.68	97.00	11.79	33.43
FGSM	ϵ =0.1	**99.96**	0.04	3.46	98.81	0.48	27.31	82.89	20.02	10.80	67.11	11.50	0.51	62.58	5.13	2.67	97.60	11.79	32.62
FGSM	ϵ =0.15	**99.96**	0.04	3.51	99.55	0.30	28.58	90.80	20.02	11.72	70.64	11.50	0.51	65.10	5.67	**2.50**	96.80	11.79	32.68
FGSM	ϵ =0.2	**99.96**	0.04	2.68	99.78	0.29	28.40	93.05	20.02	10.72	70.00	11.11	0.49	67.48	3.76	2.74	97.39	11.79	32.02
FGSM	ϵ =0.25	**99.96**	0.04	3.55	99.79	0.14	29.02	94.80	20.02	10.40	72.53	11.50	0.48	66.92	5.58	2.70	96.20	11.79	32.47
FGSM	ϵ =0.3	**99.96**	0.04	3.25	99.93	0.14	**27.21**	96.60	20.02	11.49	84.67	10.50	0.50	67.12	4.88	2.67	97.20	11.79	31.97
BIM	ϵ =0.05	**99.96**	0.04	3.27	77.22	12.23	27.47	13.80	20.02	8.96	45.36	11.44	0.54	50.70	5.08	2.66	97.20	11.79	33.13
BIM	ϵ =0.1	**99.96**	0.04	3.34	83.81	9.64	27.52	13.95	20.02	9.00	45.86	11.56	0.52	50.49	5.89	2.66	97.60	11.79	32.29
BIM	ϵ =0.15	**99.96**	0.04	3.03	88.51	5.14	27.24	16.20	20.02	9.71	47.86	11.78	0.53	51.15	5.16	2.68	96.80	11.79	32.44
BIM	ϵ =0.2	**99.96**	0.04	1.42	92.76	4.49	28.13	19.60	20.02	10.07	48.03	11.28	0.51	53.08	5.74	2.67	97.39	11.79	32.20
BIM	ϵ=0.25	**99.96**	0.04	2.62	93.27	5.00	28.64	21.70	20.02	9.76	48.97	10.94	0.51	54.88	5.08	2.66	96.20	11.79	32.01
BIM	ϵ =0.3	**99.96**	0.04	1.60	95.83	2.70	29.69	26.40	20.02	9.99	48.53	11.94	0.51	56.31	4.52	2.65	97.20	11.79	32.35
PGD	ϵ =0.05	**99.96**	0.04	1.54	91.14	5.52	28.03	21.70	20.02	9.15	50.36	11.11	0.51	51.60	5.09	2.66	97.00	11.79	32.94
PGD	ϵ =0.1	**99.96**	0.04	3.44	92.95	5.13	29.73	19.15	20.02	**8.95**	46.86	11.22	0.50	48.44	5.89	2.65	97.60	11.79	32.40
PGD	ϵ =0.15	**99.96**	0.04	1.57	95.96	2.82	29.05	18.20	20.02	9.35	46.72	11.22	5.10	49.26	5.16	2.70	96.80	11.79	32.34
PGD	ϵ = 0.2	**99.96**	0.04	3.58	97.32	2.51	30.69	16.85	20.02	9.42	50.72	11.94	0.49	50.49	5.74	2.68	97.20	11.79	32.28
PGD	ϵ =0.25	**99.96**	0.04	3.47	98.46	1.54	28.14	19.00	20.02	9.01	46.53	11.61	0.48	51.44	5.08	2.66	96.20	11.79	32.14
PGD	ϵ =0.3	**99.96**	0.04	9.80	99.29	1.03	34.45	18.80	20.02	9.01	46.61	11.39	0.49	66.98	4.51	2.65	97.20	11.79	31.94
DF	-	**99.96**	0.04	3.37	75.26	2.36	41.47	86.45	20.02	9.62	73.33	11.22	0.49	66.98	3.04	2.71	95.80	11.79	30.40
CW	κ = 0.0	**99.96**	0.04	0.90	59.57	8.33	28.51	76.85	20.02	10.03	83.39	11.22	4.70	72.82	0.67	2.75	99.00	11.79	30.82
CW	κ = 0.5	**99.96**	0.04	0.90	58.72	7.82	29.51	69.15	20.02	15.32	73.11	11.17	0.71	**76.40**	1.25	2.72	97.20	11.79	31.86
CW	κ = 1.0	**99.96**	0.04	**0.89**	89.76	4.90	27.36	70.45	20.02	9.25	85.33	11.44	0.52	49.46	4.29	2.69	**100.00**	11.79	**28.59**
CW	κ = 2.0	**99.96**	0.04	2.83	**100.00**	0.00	37.93	**100.00**	20.02	14.3	**94.36**	11.28	0.67	47.10	5.80	2.71	**100.00**	11.79	28.71

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of detection time makes SFAD a promising choice for autonomous driving systems.

D. Deep Neural Reduction (DNR) Method

The DNR method exhibits varying levels of performance across different attacks and datasets. On the TSRD dataset, it shows excellent performance against CW attacks, medium performance against BIM and PGD attacks, and poor performance against FGSM and DF attacks. Conversely, on the BTSC and GTSRB datasets, it exhibits excellent performance against FGSM, DF, and CW attacks but struggles with iterative attacks like BIM and PGD. This discrepancy can be attributed to the refinement process of adversarial examples in iterative attacks, which presents challenges for the DNR method. FPR for all datasets is consistently high, around 20%, indicating a significant number of false detections.

In terms of detection time, the DNR method falls within the middle range. It utilizes a single classifier instead of three SVM classifiers, which helps maintain a reasonable detection time. Specifically, for the TSRD dataset, the detection time is 0.1024 seconds, while for the BTSC dataset, it is 0.0864 seconds. The GTSRB dataset shows a detection time of 0.1034 seconds.

Considering the practical implications in AVs context, it is crucial to minimize detection time. While the current detection time of the DNR method is reasonable, incorporating additional classifiers could potentially increase the detection time. However, in autonomous driving scenarios, where real-time decision-making is paramount, it is essential to strike a balance between improving detection rate with multiple classifiers and ensuring that the overall detection time remains within acceptable limits. This balance allows for timely responses and meets the stringent requirements of AV systems.

In summary, the DNR method exhibits varying performance across attacks and datasets, excelling in some attacks while struggling in others. The high FPR suggests a need for further improvement in reducing false detections. Additionally, considering the unique requirements of AVs, maintaining an acceptable detection time becomes crucial, necessitating careful consideration when incorporating additional classifiers.

E. Feature Squeezing (FS) Method

In terms of performance, FS demonstrates medium effectiveness on all attacks for the TRSD and GTSRB datasets. However, on the BTSC dataset, FS performs well against all attacks except for the BIM and PGD attacks. Across all datasets, FS exhibits low FPR indicating its ability to minimize false detections.

Furthermore, the FS method showcases impressive efficiency in terms of detection time. For the TSRD dataset, the detection time is 0.0196 seconds, while for the BTSC dataset, it is 0.0257 seconds. Although slightly higher, the GTSRB dataset still presents a commendable detection time of 0.0268 seconds.

Overall, the results highlight that FS demonstrates medium effectiveness against most attacks and achieves low FPR, making it a promising method for detecting adversarial attacks in TSC. Additionally, its efficient detection time across the datasets further enhances its suitability for autonomous vehicles.

F. Adaptive Neural Reduction (ANR) Method

The ANR method demonstrates excellent performance against all attacks on all datasets. This exceptional performance can be attributed to its innovative approach of utilizing image entropy as a measure of image complexity. While ANR exhibits a low FPR on the TSRD dataset, it shows a high FPR on both the BTSC and GTSRB datasets. This discrepancy suggests that ANR may struggle to accurately classify legitimate samples in certain contexts, leading to a higher number of false detections on the BTSC and GTSRB datasets.

In terms of detection time, it is important to note that ANR has the longest processing time among all the evaluated methods. This increased detection time is primarily due to the computation time during spatial smoothing which is a key process in ANR. Specifically, for the TSRD dataset, the detection time is 0.4422 seconds, while for the BTSC dataset, it is slightly higher at 0.4711 seconds. The GTSRB dataset shows a detection time of 0.3153 seconds, which is also relatively higher compared to other methods.

Overall, the results demonstrate that the ANR method achieves excellent performance against adversarial attacks. However, the higher FPR on the BTSC and GTSRB datasets suggests a need for further improvement in accurately distinguishing between AEs and legitimate samples. Additionally, the longer detection time of ANR, attributed to its computational processes, should be taken into consideration when evaluating its suitability for autonomous driving systems.

G. Summary on Detection Rate and Detection Time

In terms of detection rate, the ANR method consistently outperforms all other methods on all datasets, as shown in the last part of Figure 3. It is followed by NSS and LID, which also demonstrate strong performance. SFAD, FS, and DNR rank lower in terms of detection rate. However, it is important to consider the higher FPR associated with ANR compared to NSS and LID. Among NSS and LID, LID exhibits a higher FPR. As a result, NSS emerges as the appropriate method, striking a balance between detection rate and FPR.

When considering detection time, SFAD demonstrates the lowest processing time among all evaluated methods, as shown in the last part of Figure 4, followed by NSS and FS. DNR, LID, and ANR rank higher in terms of detection time.

For critical applications such as autonomous vehicles, where both detection rate and detection time are crucial factors, NSS becomes the top contender. It offers a favorable combination of high detection rate, lower FPR, and reasonable detection time.

In summary, considering the collective factors of detection rate, FPR, and detection time, NSS emerges as the most suitable and favorable method among all the evaluated methods on all datasets. While the ANR method achieves the

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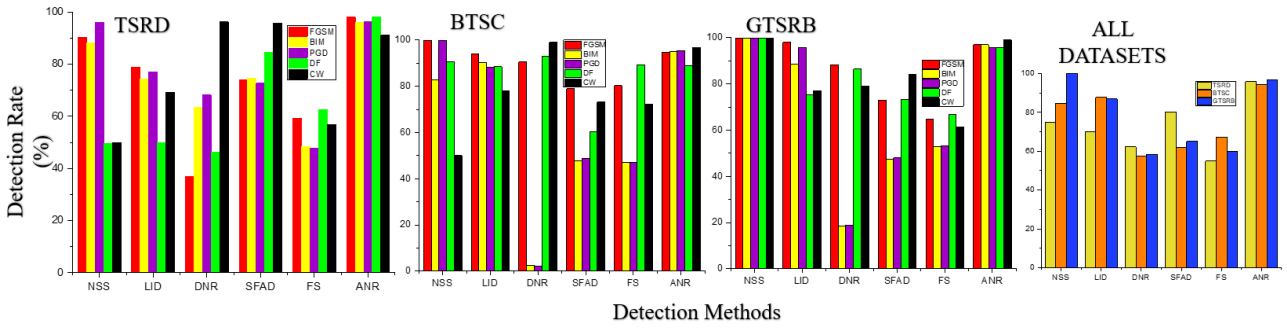


Fig. 3. Performance comparison of the six detection methods against white-box attack based on Detection rate.

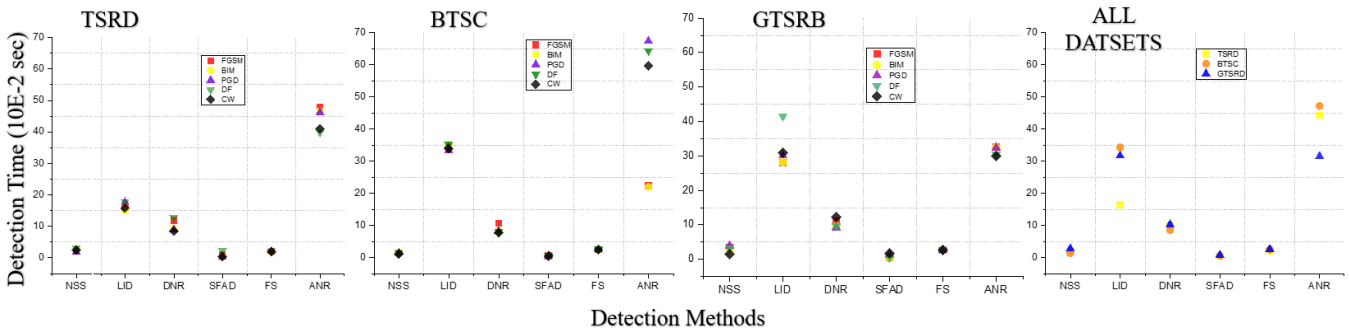


Fig. 4. Performance comparison of the six detection methods against white-box attack based on Detection time.

highest detection rate, it also exhibits a higher FPR. SFAD demonstrates the lowest detection time, but NSS strikes a balance between detection rate, FPR, and detection time, making it the optimal choice for autonomous vehicles.

VI. Future Work and Conclusion

This section summarizes the findings of our study and discusses their implications for TSC in AVs. It also highlights potential areas for future research and development, focusing on key aspects that require further investigation and improvement.

A. Future Work

1) Multimodal Sensing Approaches: Our study highlights the effectiveness of various adversarial detection methods for TSC in AVs, yet there is considerable potential for further advancements. Recent innovations in sensing technologies, such as the vision-based soft tactile sensing system in ViART [15], demonstrate the potential for integrating advanced modalities. Future research should explore multimodal sensing approaches, combining visual and tactile data, to enhance the robustness of adversarial detection methods in TSC. This integration can ensure timely and accurate responses to adversarial threats, ultimately improving the safety and reliability of autonomous driving systems.
2) Real-time intelligent detection strategies: This study emphasizes the potential for further enhancement in adversarial detection methods for TSC. Approaches like AutoDetect [16] highlight the importance of real-time intelligent detection systems, which offer real-time processing and sensor failure identification. Future research should focus on integrating these advanced detection technologies to strengthen the robustness and reliability of TSC methods for AVs. This approach will ensure rapid and precise responses to adversarial threats, significantly enhancing the safety and dependability of autonomous vehicles.
3) Cloud-Based IDS strategies: Strategies such as RCA-IDS demonstrate the advantages of real-time cloud-based Intrusion Detection Systems (IDS) [51], offering capabilities like real-time sensor data monitoring and attack detection. Future research should focus on integrating these advanced cloud-based IDS approaches to improve the robustness and scalability of adversarial detection methods in TSC. This integration will ensure timely and accurate responses to adversarial threats, enhancing the safety and reliability of autonomous driving systems.

By addressing these areas of future research, we can advance the field of adversarial detection in TSC for AVs, leading to more robust, and efficient methods that ensure the safety and reliability of autonomous transportation systems.

B. Conclusion

Our comparative study highlights the resilience of detection methods against sophisticated adversarial attacks, emphasizing their potential to enhance crucial safety protocols for autonomous navigation. Through the evaluation of various detection methods on different datasets, we have gained valuable insights into their performance and trade-offs.

Among the evaluated methods, the ANR method achieves the highest detection rate across all datasets, although it

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comes with a higher false positive rate. On the other hand, SFAD demonstrates the lowest detection time on all datasets, indicating its efficiency. Meanwhile, the NSS method strikes a balance between detection rate, false positive rate, and detection time, making it a suitable choice.

These findings provide valuable guidance for selecting detection methods based on specific requirements and preferences on TSC. By leveraging adaptive learning techniques with GAN and prioritizing time-efficient methods, we can significantly enhance the safety and reliability of autonomous transportation systems. This research contributes to the advancement of adversarial attack detection in TSC for AVs, paving the way for a future where AVs can operate securely and reliably in real-world environments, effectively mitigating the risks associated with adversarial attacks.

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